LEVERAGING ARTIFICIAL INTELLIGENCE FOR PREDICTIVE CYBERSECURITY THREAT INTELLIGENCE

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Declaration

I hereby assert that this dissertation, entitled Leveraging Artificial Intelligence in Predictive Cybersecurity Threat Intelligence, is my original work, conducted in partial fulfilment of the requirements for my degree. To the best of my knowledge, this has not been presented, in whole or in part, toward any other academic qualification at this or any other institution. All the information, references and contributions of others have been explicitly indicated and cited accordingly to the academic standard.

Abstract

The swift rise in the complexity and frequency of cyberattacks has brought to light how ineffective conventional security tools are when combating rare and highly dangerous attacks this is because traditional security tools usually cannot detect rare but dangerous attacks. This paper discusses how to apply artificial intelligence (AI) and machine learning (ML) in predictive cybersecurity to detect minority-class attacks, including infiltration and web attacks as well as advanced persistent threats (APTs). The study uses SMOTE to overcome the problem of class imbalance on the CIC-IDS2018 dataset by resorting to preprocessing and feature selection techniques and resampling strategies. Four models: Random Forest, Support Vector Machine (SVM), XGBoost, and Logistic Regression were trained and compared using precision, recall, F1-score, and confusion matrices.

The findings revealed that the ensemble models (Random Forest, and XGBoost) showed off-the-chart accuracy, recall, and F1 -score results, a signaling strong performance with the benign and malicious traffic. Although SVM and Logistic Regression could have done slightly worse at benign traffic classification, it was very high at malicious traffic, ensuring that critical threats were detected. The results support the conclusion that the use of AI-enabled models leads to a significant enhancement of cybersecurity as false negative rates are minimised, and predictive intelligence is maximised when the models are optimised and balanced. The paper ends with a recommendation to test models before they are introduced to real-trial traffic, combine both continuous and discrete resampling methods, incorporate explainability tools like SHAP and LIME, and to modify sources regularly in order to reflect on the emerging threats.

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Chapter 1: Introduction

1.1 Background of the Study

The digital revolution of the contemporary world has opened up inexhaustible possibilities for increasing growth, innovation, and convenience in the economy. Nevertheless, such a change has resulted in serious vulnerabilities, causing the emergence of more complex cyber threats. The world has witnessed a substantial rise in the frequency and sophistication of cyberattacks over the past decade (Mahfuri *et al.*, 2024). Ransomware, advanced persistent threats (APTs), zero-day exploits, and phishing have all posed significant threats to critical infrastructures, government systems, businesses, and individual users, robbing them of revenue, exposing sensitive data, and creating vulnerabilities that need to be addressed (Baksi *et al.*, 2023).

Traditional security tools, such as firewalls and signature-based intrusion detection systems, are struggling to keep pace with these evolving threats (Lyu *et al.*, 2024). Such traditional systems are quite reactive, hence not valuable for unknown attacks, as they rely on known attack signatures or behavioural rules. The high-level sophistication of threat actors, along with the large volume of network traffic that requires real-time handling, is also the reason why it is becoming increasingly complex to detect malicious activities in time before significant damage can be done manually through security systems (Landauer *et al.*, 2022).

The most apparent solution to the raised concerns is infamous artificial intelligence (AI), particularly the machine learning (ML) algorithm for proactive threat detection. Big data can be learned using these intelligent systems, which can detect complex patterns and make real-time predictions. AI finds applications in the cybersecurity sector to identify abnormalities, track intrusions, and predict potential attacks that may occur. However, the major problem is the high imbalance within cybersecurity datasets, where attacks by the minority class are most likely to be underrepresented. Thus, even more advanced ML algorithms may struggle to identify the fact that rare but highly dangerous attacks occurred (Mirsadeghi *et al.*, 2023).

One of the standard benchmarks in this field is the UNB dataset (UNB, 2018), which combines various types of attacks with realistic simulated traffic. Although such data can be used to construct a comprehensive framework for evaluating intrusion detection models, there is an imbalance in representing the reality of classifying minority-class threats in a real-world problem. Such an imbalance usually leads to the failure in identifying rare attacks because most machine learning algorithms tend to focus on the majority of classes at the expense of the minority class. To tackle this problem, sophisticated approaches, including Such methods as

the Synthetic Minority Over-sampling Technique (SMOTE) and the Adaptive Synthetic Sampling (ADASYN), should be used along with the improvement of the processes of model training and evaluation (Karatas *et al.*, 2020).

1.2 Overview of the Study

The research aims to investigate how artificial intelligence can be used to improve predictive cybersecurity threat intelligence and identify minority-class cyber threats through its application. The goal of this effort is to develop a machine learning system capable of detecting rare attacks, such as infiltration, web, or APTs, with high predictive confidence in an unbalanced dataset (UNB, 2018). The authors will utilise three popular machine learning algorithms: Random Forest, Support Vector Machines (SVM), and XGBoost (Khan *et al.*, 2024). These models have been selected based on their efficiency in handling classification tasks and their applications in cybersecurity research.

These are very complex resampling algorithms that have been employed to balance the dataset, thus enhancing model performance, such as SMOTE and ADASYN. This type of approach will provide synthesised instances of minority-class attacks, enabling models to be trained more effectively to recognise their characteristics. Moreover, the models are optimised concerning hyperparameters using Grid Search and Bayesian Optimisation (Albasheer *et al.*, 2024). This entails adopting optimal settings in each model, thereby enhancing accuracy and recall rates.

It is evaluated based on major metric scales, including precision, recall, F1-score, and false negatives. Special focus is given to the possibility of achieving a recall rate of at least 90 percent for minority-class threats and false negative results, with no more than a 30 percentage point increase over the baseline models (Al-Shehari *et al.*, 2024). These benchmarks are significant because any false negatives in the case of an attack can be fatal to an organisation.

The entire experiment is organised, beginning with data preprocessing and exploratory data analysis (EDA), and continues with resampling and feature selection, training, validation, and evaluation of the models. The consideration of interpretability is also incorporated, and the approaches of explaining predictions, including SHAP (Shapley Additive Explanations) and LIME, are implemented (Birahim *et al.*, 2025). This provides clarity that not only is the end framework accurate, but it is also open and understandable, thus enabling the framework to run smoothly in the practical world, including Security Operations Centers (SOCs).

1.3 Problem Statement

Cyber-attacks are becoming increasingly complex, and organisations are being exposed to advanced threats, as conventional detection systems can no longer keep up with them. Despite the strong arsenal of tools created by artificial intelligence to optimise the process of detecting threats, to a significant degree, the search models used cannot operate as effectively due to the obvious imbalance of the dataset in the field of cybersecurity. However, the least common yet the most dangerous and destructive signals of cyberattacks are those posed by representatives of minority classes (Altalhan *et al.*, 2025). Traditional machine learning models are commonly biased towards the majority classes, especially when faced with rare attacks, resulting in a high false negative rate.

Although there are labeled datasets (such as UNB, 2018) that include several types of real attack scenarios, they are also highly imbalanced, posing a formidable obstacle to precise detection. Next, there is a demand to develop a predictive threat detection system that can recognise attacks on minority classes with a high level of recall and low false-negative rates. The study aims to bridge this significant gap by introducing an optimised and AI-based detection framework that can ultimately enable improved predictive cybersecurity within an operating environment.

1.4 Research Aim

The proposed research study will develop an optimised machine learning framework that enhances predictive cybersecurity threat intelligence, with a particular focus on improving the sensitivity of imbalanced datasets to detect minority-class attacks (e.g., ransomware, APTs). The potential of this study lies in bridging the gap between the performance of theoretical models and practicality, ensuring that cybersecurity systems are proactive enough in recognising emergent threats with high accuracy and a low number of false negatives. This is possible by utilising advanced AI algorithms.

1.5 Research Objective

1. To preprocess and analyse the CIC-IDS2018 dataset, extracting meaningful features and addressing class imbalance using resampling methods such as SMOTE and ADASYN.
2. In order to hyperparameter tune machine learning models (Random Forest, SVM, and XGBoost) to maximise their performance, and guarantee the model is effective in identifying rare cyber threats.

3. To compare and contrast the models based on key measurements, including recall (>= 90 percent), precision, and F1-score, with a focus on a minimum of 30 percent reduction of false negatives when compared to standard methods.
4. To find the most efficient model to detect the threats related to the minority classes and deliver the insights that can be employed in real-life scenarios when it comes to implementing the AI-based cybersecurity defenses.

1.6 Research Questions

1. Will the optimised machine learning models, Random Forest, SVM, and XGBoost, achieve a recall of 90% in detecting minority-class cyber threats based on the (UNB, 2018) dataset?
2. Will resampling methods such as SMOTE and ADASYN considerably decrease false negative rates, especially in rare attacks like infiltration and web attacks?
3. How can the level of matching between detection and computational efficiency in predictive cybersecurity be found using Random Forest, SVM, and XGBoost models?
4. What are possible solutions and methods that explainability tools (for example, SHAP, LIME) can do to make AI-powered cybersecurity systems appear to be more trustworthy and interpretable?

1.7 Hypotheses

- H1: When resampling is taken as an optimisation method, ML models will demonstrate a 90-100% recall of the minority-class threats present in the (UNB, 2018) dataset.
- H2: SMOTE or ADASYN application will lead to a reduction in the false negatives of at least 30 percent, compared to the baseline models found on the unequally balanced dataset (Popoola *et al.*, 2025).
- H3: XGBoost will be the best in comparison with Random Forest and SVM in the selected models in terms of the F1-score and recall, especially when the classes are imbalanced.
- H4: The integration of tools to interpret the model will offer valuable insights into the model predictions, making them more applicable to real-world cybersecurity systems.

1.8 Rationale of the Study

The justification for this research is related to the need to enhance cybersecurity protection through predictive and intelligent threat detection (Al-Quayed *et al.*, 2024). The vast majority of existing systems are designed to respond to known threats, rather than new ones that have

not yet been encountered. With an increased amount of traffic on the network and the sophistication of the attack strategies in place, a quick and effective automated response is virtually impossible by manual or rule-based systems.

Machine learning holds great promise for addressing this problem, and its effectiveness strongly relies on its ability to deal with class imbalance, which is highly prevalent in threat data. Even advanced models cannot alert to mere occasional attacks unless their data distribution is rectified. With the application of SMOTE and ADASYN, the training data becomes more representative, and the models are trained on both frequent and infrequent attacks (Abbasi *et al.*, 2025).

Besides, security teams need not just models that are accurate but also ones that can be understood and explained. The decision-making process of black-box models can be very adequate, but it does not command trust as long as the decision-making process is opaque (Gaspar *et al.*, 2024). This paper will address this concern by incorporating the interpretation of models into the research design. Due to this, the proposed framework not only ensures technical performance but also aligns with the practical needs of deployment in real-life settings.

1.9 Scope of the Study

The present paper is devoted to predictive identification of the threat of cybersecurity with the help of machine learning and artificial intelligence. The main dataset affected is UNB (2018), one of the most popular benchmarks of intrusion detection. In this data set, it is the case of minority-class threats which are the most significant ones to the study such as infiltration, web attacks, APTs and many others as they are unique and complex.

Only the following machine learning algorithms are studied: Random Forest, Support Vector Machines and XGBoost. It is because they proved effective in classifying problems, even those that had imbalanced data (Shtayat *et al.*, 2023). Deep learning models, neural networks, and ensemble stacking are excluded from the study due to time and resource limitations, as well as the need for explainability.

The studies only involve offline analysis, that is, testing the models using already available datasets, rather than in a real-time setting. Nevertheless, the last framework is destined to be built with the possibility of integration in real-time systems. Additionally, ethical aspects, such as data confidentiality, fairness of the model, and transparency, are considered to be compliant with legal and ethical frameworks.

1.10 Dissertation Structure

The structure of this dissertation. The research process is organised into five chapters, each of which addresses a particular aspect of the research process.

- Chapter 1: Introduction - Gives the background, the problem being addressed, goals, and the research hypotheses and a survey of the research.
- Chapter 2: Literature Review - describes the existing studies concerning AI in cybersecurity, class imbalance, and machine learning models to detect threats.
- Chapter 3: Methodology - Outlines the research design, data acquisition, preprocessing procedures, algorithms, evaluation criteria and interpretability option.
- Chapter 4 Results and Analysis- Provides the results of the experiment, the comparison of models, performance, additional important features, and error analysis.
- Chapter 5: Conclusion and Recommendations- Contains a summary of key findings, contributions, limitations and recommendations on research and practical application in the future.

Each chapter builds on the previous one to provide a comprehensive understanding of how artificial intelligence can be leveraged for predictive cybersecurity threat intelligence. The outcome is expected to contribute to both academic research and practical solutions for cyber defence systems.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The rate at which cyber threats have been growing in number and complexity has led to increased demand for the development of intelligent systems, which can anticipate and stop attacks before they result in serious damage. Conventional rule-based applications cannot typically detect advanced threats, in particular, when the datasets are unbalanced and/or do not provide insight in time. Current research on artificial intelligence (AI) and machine learning (ML) has provided a prospective opportunity to upgrade cybersecurity through precise identification of rare and stealthy attacks (Kavitha and Thejas, 2024). Predictive systems of threat intelligence based on optimised ML models are emerging as a central approach to enhancing digital defence infrastructure in any industry.

This chapter addresses the base ideas, techniques and obstacles that characterise the development of AI-enabled cybersecurity infrastructures. It examines works available on the models of threat detection, the class imbalance problem and the incorporation of resampling methods such as SMOTE and ADASYN. It includes the feature extraction, hyperparameter tuning, and metrics of evaluation that help to increase model stability in the literature as well (Yang *et al*., 2024). In addition, issues of ethical concerns, interpretability of the models, and strategies of real-time deployment are discussed to give a good understanding of the landscape. This review puts a clear context to the development of strong, proactive, and practical threat identification systems as laid out in this chapter, which aligns with the goals of the research and makes significant contributions to both theoretical and practical cybersecurity.

Figure 1: AI in CyberSecurity

(Source: Data Science Dojo, 2023)

2.2 Preprocessing and Handling Imbalanced Data using SMOTE & ADASYN

According to Albasheer *et al.* (2024), the CIC-IDS2018 has been well heralded due to its realistic nature in network traffic and attack simulation. It also consists of multiple kinds of cyber threats, which makes it appropriate to assess the ability of the machine learning models to detect intrusions (Albasheer *et al.,* 2024). The disadvantage of this dataset is, however, one of its biggest problems: the imbalance between attack and normal traffic. Attacks comprise only a small percentage of total traffic in most scenarios in the world and, as a consequence, they become difficult to spot. Machine learning models can do very poorly on these underrepresented classes. Hence, prior to developing any predictive model, it becomes necessary to conduct preprocessing tasks in order to clean and prepare the information. These are the manipulation of missing values, the elimination of duplicates and the standardisation of what data should look like (Karatas *et al.*, 2020). Another important part of this stage is feature extraction, because it assists in the identification of the most crucial fields describing suspicious or safe behaviour.

From the viewpoint of Mirsadeghi *et al.* (2023), after cleaning and preparation of data, the next issue is the class imbalance. The majority of conventional machine learning algorithms assume balanced classes and hence tend to take the majority class at high rates. It causes more false negatives, which is extremely dangerous in the situation of cybersecurity (Mirsadeghi *et al.,* 2023). Resampling methods are employed in order to address this problem. The synthetic minority over-sampling technique (SMOTE) fills the minority class with new samples that are created by interpolating among examples. It assists the model in learning using a different distribution of classes, which is more balanced. ADASYN (Adaptive Synthetic Sampling) on the one hand generates new samples u yet also corrects the amount of samples depending on the learning difficulty (Altalhan *et al.*, 2025). This implies that ADASYN is more concerned with the areas where the model does not work well; hence, this makes it more efficient when it comes to picking up intricate patterns in the data.

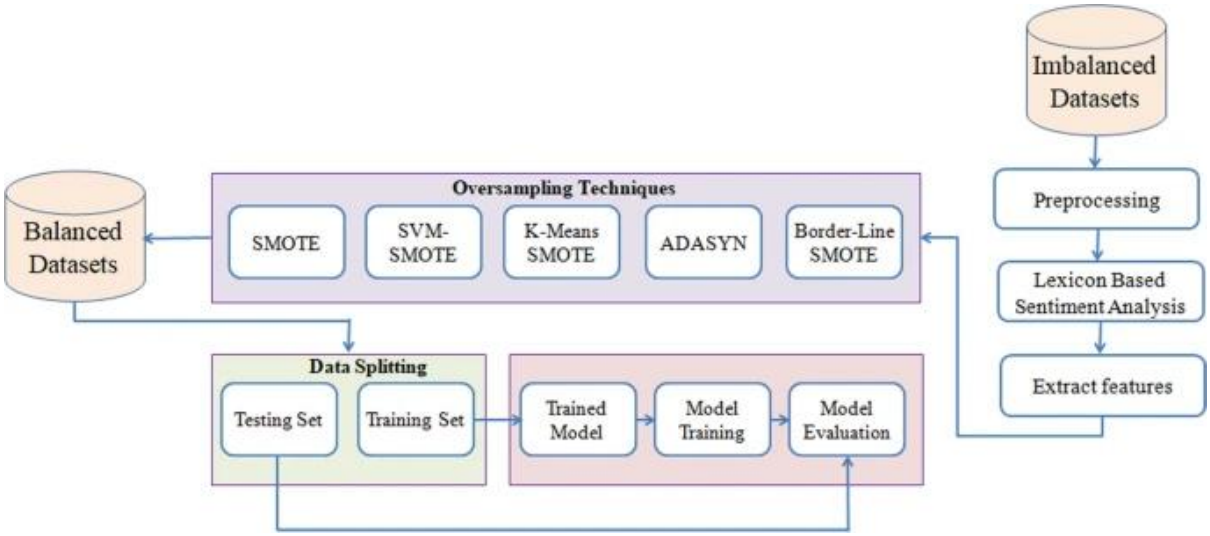

Figure 2: Data Preprocessing in Cybersecurity

(Source: Mujahid *et al*., 2024)

As per the findings of Alamri and Ykhlef (2024), resampling techniques have to be applied cautiously. In the event that the oversampling is not balanced, then noise may be added, and overfitting may occur. So, prior to resampling, feature selection needs to be undertaken (Alamri and Ykhlef, 2024). The feature selection simplifies the model, motive and precision because it decreases the number of input variables. Correlation analysis or recursive feature elimination are some of the methods that can assist in finding the most meaningful features. After the resampling step, features are used to train the models. The models will hopefully perform better with improved identification of common and rare forms of attacks since they are trained on a cleaner and more balanced piece of data.

According to Popoola *et al.* (2025), pre-processing, feature selection, and resampling have been proven to work positively in the new trends of studies. Such measures enhance the recall of threats in the minority-class class without driving false positives up to unmanageable numbers. The SMOTE and ADASYN can be applied in the context of the CIC-IDS2018 dataset, in which attacks such as infiltration, botnet, and web attacks do not occur very frequently and enhance the accuracy of detection in those cases significantly (Popoola *et al.,* 2025). The trained models on this processed data are more sensitive to rare attacks, which are necessary for real-time threat prediction. Such gains not only enhance the performance but also the trust in the system as well, particularly in cases where a failure in detecting a single attack may cause critical outcomes.

On the basis of Kumar and Kumari (2025), the final action on this path should include the confirmation of the effects of preprocessing and resampling on the performance of the model. The performance is measured with precision, recall, F1-score and false negatives. Remarkable emphasis is placed on recall since this indicator determines how accurately the model predicts real attacks. When recalls are low, few attacks are detected, and that is dangerous. They practice better performance by simplifying those models that were trained by SMOTE and ADASYN. Although accuracy may also be slightly reduced as a result of creating synthetic data, the increase in recall may easily offset this shortcoming. Finally, preprocessing and resampling are also important aspects in constructing good machine learning systems when dealing with cybersecurity and, more importantly, when dealing with class-imbalance datasets such as CIC-IDS2018.

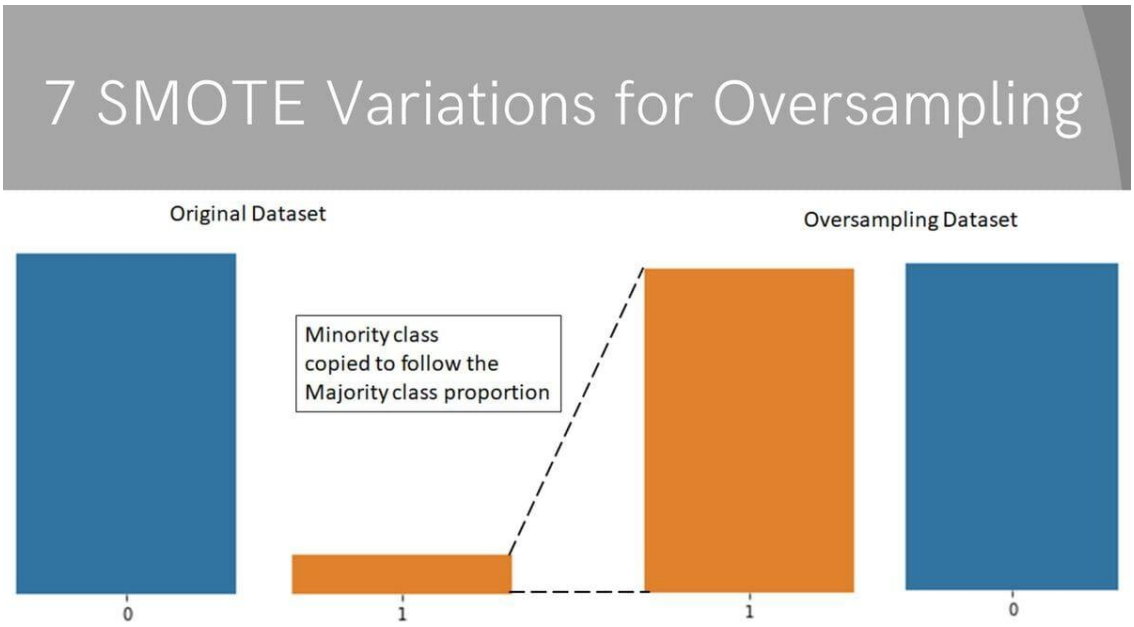

Figure 3: SMOTE Variations for Oversampling

(Source: KDnuggets, 2023)

2.3 Hyperparameter Tuning of RF, SVM, and XGBoost for Rare Threat Detection

According to Sannigrahi and Thandeeswaran (2024), it is established that machine learning models fare best after their hyperparameters (the internal settings) are adjusted properly. These hyperparameters are the parameters that regulate the learning of a model on the data, as well as the decision-making of the model. Unless adjusted, the model can fail to perform as desired, particularly on matters such as cybersecurity, which are complex. In this regard, the Random Forest, Support Vector Machine (SVM) and XGBoost models need to be tuned appropriately to be functional. All these models are accompanied by parameters. As an example, Random

Forests have a number of trees that can influence the effectiveness of threat detection, and the learning rate of XGBoost can influence it, as well (Tufail *et al.*, 2025). In the absence of tuning, the models can either be overfit or fail to learn crucial trends in the data.

From the viewpoint of Sahu *et al.* (2024), hyperparameter tuning involves experimenting with various combinations of values to produce the optimum set-up of a model. There are many ways to do this process, for example, Grid search or Random search. Grid Search searches through all of the combinations in a given range, whereas Random Search selects random values in the same range. Such practices are common since they are straightforward and easy to apply (Sahu *et al.,* 2024). They may, however, be time-consuming when many parameters have to be tried. To overcome this, higher-order techniques such as Bayesian Optimisation have been proposed. These optimise quickly by performing smart guessing to arrive at the best values. Regardless of what technique is applied, tuning is one of the essential components in developing effective machine learning models when rare cyber threats need to be discovered.

As per the findings of Dasari and Kaluri (2024), tuning may differently affect each machine learning model. Let us take an example, Random Forest works better when the depth of a tree and the number of trees in the forest are well-tuned. Otherwise, the model can fail to detect meaningful patterns in case the trees are shallow. When they are too deep, then the model can learn noise rather than real signals (Dasari and Kaluri, 2024). The type of kernel and the value of regularisation are quite crucial in the case of SVM. The proper selection of the kernel will enable the model to distinguish between normal and attack traffic. XGBoost needs the adjustment of learning rate, boosting rounds, and tree depth. Having more rounds on a lower learning rate may present better results, but a lot more computer time is needed (Alekseeva *et al.*, 2021). Yet, such changes, even though small, can make a huge impact on the detection of rare threats.

According to Kandhro *et al.* (2023), less frequent cyber threats, such as infiltration or advanced persistent threats, are more difficult to identify since they are less visible within the information. A well-adjusted model has a higher chance of identifying such hidden patterns. The hyperparameters should therefore be selected wisely in order to allow the model to concentrate even on the diminutive signals. This decreases prompting missed attacks and enhances the dependability of the system. And it minimises the number of false alarms, which are typical in models which are not tuned well (Kandhro *et al.*, 2023). Such a ratio of sensitivity

and precision is particularly significant to cybersecurity, where missing a threat may have severe repercussions.

From the viewpoint of Raza *et al.* (2023), when models are tuned, they have to be evaluated with metrics. These are Accuracy, precision, recall and F1-score. The most useful of them to the cybersecurity area is recollection, which reflects the number of real threats being identified by the model. Good memory suggests that the chances of missed attacks are reduced (Raza et al., 2023). Tuning can be employed to strive towards maximising this value because there is a need to make the model more biased towards learning the correct patterns. In the case of rare threats, a marginal increase in recall can make the system much safer. The testing ought to be conducted in an attentive manner so that the model is not tested solely on the training data, but it is effective even on unseen new data (Sudyana *et al.*, 2024).

Overall, hyperparameter tuning is not a technical process; it is a mandatory process to produce solid cybersecurity models. Even such a developed model as Random Forest, SVM, and XGBoost could not identify the rarest and most dangerous cyber threats in case of improper tuning. The process of tuning by dedicating some time will make the models more accurate, fairer, and ready to face the real world (Mohammed *et al.*, 2024). This will make the end result a smart system that is also reliable, which provides a high level of protection against any form of cyber threat.

2.4 Model Comparison Using Recall, Precision, F1-Score & False Negative Reduction

According to Al-Shehari *et al.* (2024), the effectiveness of the machine learning models is rather significant in the sphere of cybersecurity. This aids in determining the accuracy of each model to identify the threats and, more so, the infrequent ones. Three of such evaluation criteria are normally used, namely: precision, recall and F1-score. They are used in order to analyse the ability of the model to identify threats properly and reduce the number of false mistakes (Al-Shehari *et al.*, 2024). Recall reports the number of real attacks that were observed by the model. Accuracy becomes the number of expected threats that were accurate. F1-score has a compromise between the two. These metrics prove to be very useful when comparing models like Random Forest, SVM, and XGBoost, in particular, when dealing with an imbalanced dataset in which the majority of traffic may be normal and only a minor fraction is malicious (Pavithra and Vikas, 2024).

From the findings of Li (2024), the individual performance metrics on these models show that models tend to give varied results on these metrics. A model that has high precision can fail to

capture numerous attacks when the recall is low. To take an example, even the most accurate models to label the safest traffic can still not identify real threats. In cybersecurity, it cannot afford to miss an attack because this is just as bad as creating an unnecessary alarm (Li, 2024). That is why one may speak about recall as the most significant metric. It should have a high recall value of about 90 per cent or higher so that the majority of the threats could be detected. One should not simply concentrate on correctness since this will be misleading most of the time when data is normal (Kazmi *et al.*, 2025). Rather than looking at recall, precision, and F1-score, one can get a better picture of the model working.

From the viewpoint of Alketbi and Mehmood (2025), another major consideration in framing comparisons of the models is the reduction of false negatives. False negative refers to the fact that a real threat is disregarded or not spotted by the model. It may have severe consequences, in particular in networks where a single misinterpreted attack can result in huge losses (Alketbi and Mehmood, 2025). Unoptimised models that are not trained appropriately when it comes to machine learning usually have high false negative rates. Thus, when comparing models to each other, it should be worth examining how they approach false negatives. The goal will be to lessen them by at least 30 per cent relative to normal models. Such models, as XGBoost, have proved to be more efficient in reducing false negatives in contrast to others with proper tuning.

According to Muhammad *et al.* (2025), depending on the kind of data and the complexity of the data, modelling reacts differently in each model. An example is Random Forest, which can run on big data and better handle missing values. It also entails improved outcomes in any situation where the intention is to discourage false negatives (Muhammad *et al.*, 2025). SVM, however, is effective when there is a presence of clean margins among the classes, but it is poor in ensuring high recall. XGBoost is characterised by high generalisability in terms of all three metrics. It tends to provide higher accuracy and F1-score due to the way it constructs its trees and prioritises error-learning (Upadhyay *et al.*, 2020). An assessment of these models when they are applied to normal datasets aids in clarifying where these models triumph and where they fail miserably.

When comparing the last two models, one should not consider one of the metrics but rather look at all three of them altogether. A model can have a great recall and a very low precision, thus providing a large number of false alarms. On the same note, a high precision, low recall model will fail to detect sufficient threats. F1-score can also be used in the maintenance of balance. When comparing it with this, one must come alongside with the same data, same test

conditions and resampling techniques, and this must be fair (Ali *et al.*, 2024). This will enable a fair verdict to be made regarding the best model when dealing with actual cybersecurity scenarios, especially when attempting to identify the occurrence of lesser threats, but grave when discovered.

2.5 Identifying the Best Model for Real-World Minority Threat Detection

According to Azam *et al.* (2023), identifying cyber threats belonging to minority classes forms a significant challenge in the development of reliable intelligent cybersecurity systems. Threats to the minority class include the less frequent but highly potentially harmful ones, including advanced persistent threats, web attacks, or infiltration. Such threats tend to be ignored as their presence is minimal in relation to general network traffic (Azam *et al.*, 2023). Consequently, most machine learning models might neglect them and pay more attention to the majority. Here, the need to identify an effective model arises. The effective model should be in a position to identify not only the prevalent attacks but also the uncommon threats without missing them.

From the findings of Jiyad *et al.* (2024), Support Vector Machines, Random Forest, and XGBoost are some of the popular ones to perform cybersecurity tasks. And all of these models have behaved differently when applied to unbalanced databases. Random Forest is effective in its operation since it leverages the utilisation of multiple decision trees and the joint prediction of the trees. The model tends to perform well with noisy data, and powerful results occur with a large dataset (Jiyad *et al.*, 2024). SVM can be very effective in classification, but it can also have problems when the distribution of data classes is not distinct or in the presence of a strongly unbalanced dataset. XGBoost is unique because it first of all learns from the mistakes that it has made, and it gives additional focus to hard samples, which makes it more inclined towards catching a minority class of threats.

According to Neupane *et al.* (2022), the real worth of a model not only depends on whether it can detect or not, but it also depends on the relevance of its findings in real-life scenarios. Time and proper accuracy are important in a real security operation. A model that identifies the threats in early stages, with the convincing reasons why it has alerted a specific activity, would aid in the rendition of faster decision-making by cybersecurity teams (Neupane *et al.*, 2022). So, one hopes that the selected model should not only do well on technical measures but also provide informative observations. This involves the description of major features that assisted in the identification of the threat, in order to enable the security team to learn about them and take some appropriate actions. The advantage of models such as XGBoost and Random Forest

is that they provide feature importance scores that show which input is the most helpful for the prediction.

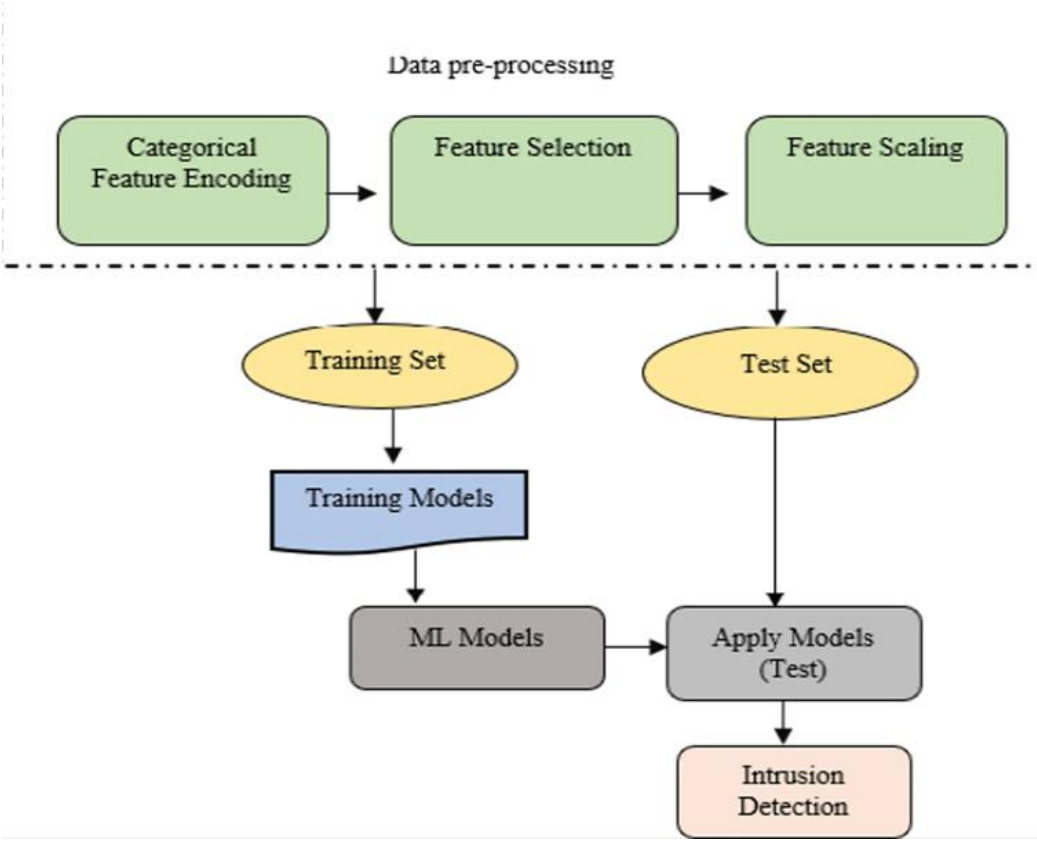

Figure 4: Feature selection pipeline and interpretability

(Source: Disha and Waheed, 2022)

In any real-world scenario, models have to be deployable and maintainable. The next solution must have the promise of processing the new data fast and changing when threat patterns transform. There must be a compromise between fast and proper. The deep learning techniques could be slower and explanatory (Yu *et al.*, 2024). Though they could deliver great accuracy, it could be difficult. Models such as XGBoost are instead a good tradeoff. They are quick, precise and slightly interpretable, and this quality makes them a powerful alternative to operational applications. The integration of model performance and real-time usefulness means that the intelligence obtained is not theoretical but can be utilised in order to actively combat the threat and thwart it.

2.6 Literature Gap

The main problem in research in the domain of predictive cybersecurity is the emphasis on the detection of popular attacks on the basis of balanced data, which is inapplicable to the real

conditions in the network. Numerous studies do not pay attention to the problem of detecting minority-class threats like infiltration and advanced persistent threats, which are extremely harmful, but their occurrence is uncommon. Moreover, although many models have high accuracy, they have poor recall on minority classes, and their applicability in real-time scenarios is poor (Govea *et al.*, 2025). In addition to that, there is an absence of directions to merge interpretability and model efficiency. Therefore, there is an evident divide between the establishment of AI models that are effective in detecting rare threats and the ones that can be implemented into the real world.

2.7 Conceptual Framework

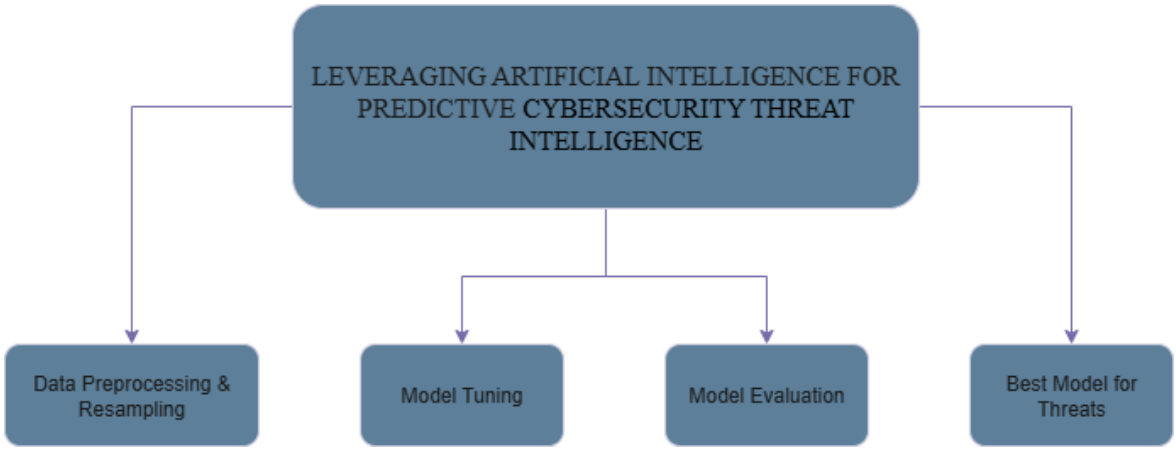

Figure 5: Conceptual Framework Diagram

(Source: Self-Developed)

2.8 Chapter Summary

The use of artificial intelligence in predictive cybersecurity, especially in the detection of rare threats by machine learning, is discussed in this chapter. It investigated why it is critical to preprocess, choose features, and manage class imbalance through algorithms such as SMOTE and ADASYN. The chapter compared other models like Random Forest, SVM, and XGBoost with higher detection rates and lower false negatives, based on the parameters recall, precision, and F1-score (Al-Shurbaji *et al.,* 2025). It also talked about how to choose an efficient model to be used in real life, based on accuracy, speed, and interpretability. In general, the review not only pointed out challenges but also revealed the necessity of applying feasible, moderate, and smart solutions regarding the security threat detection systems.

Chapter 3: Methodology

3.1 Introduction

RISP-DM (Cross-Industry Standard Process for Data Mining) is yet another rigorous and standard approach to regulating data mining and machine learning endeavours. RISP-DM was created towards the end of the 1990s and presents a six-phase strategy for data scientists to systematically analyse large-scale projects that otherwise get complicated, resulting in spending time and money as well. The first step in the process is Business Understanding, where we define project objectives and map the business problem into a data mining question (Wiemer *et al.,* 2019). Next comes Data Understanding, where the analysts collect and explore data to evaluate its quality, relevance, and insights they could potentially discover through exploratory data analysis (EDA). The Data Pre-processing phase cleans, transforms and prepares the data to model it by dealing with missing values, scaling and selecting features, among others.

At the time of Modelling, appropriate machine learning algorithms are selected, used and refined to check their fit within the data. The Evaluation phase is where the chosen model gets thoroughly tested to determine if it meets the business's objectives, and performance can be tweaked if necessary. The very last Deployment, which refers to the model being implemented in a production-like setting where it can be used for its insights or predictions (Martínez-Plumed *et al.,* 2019). The iterative nature of CRISP-DM enables continuous model refinement, making it application-agnostic and guaranteeing that the constantly updated model stays in sync with business objectives over the course of a project.

3.2 CRISP-DM

3.2.1 Business Understanding

Cyberattacks are becoming more recurring and risky, affecting people, companies, and governments across the world. Core security tools are not sufficient to identify new and hidden risks or threats. Smart systems are needed in businesses which can detect and stop cyber threats in advance. The study aims to develop a solution using different machine learning approaches which helps in detecting and predicting cyberattacks from the historical data (Charbuty and Abdulazeez, 2021). With better threat detection, the companies will be able to preserve their data, minimise financial losses, and gain trust among customers. It is aimed at assisting companies in remaining secure in the modern digital environment with the help of the latest AI-powered cybersecurity solutions.

3.2.2 Data Understanding

The dataset has 1,048,575 rows and 80 columns. The data is collected to identify network-based attacks. Every row contains data related to destination, source port, packet statistics, flag counts, protocols, and flow duration. The dataset has both categorical and numerical data, along with a target column which is labelled as Label. The dataset has timing attributes such as Timestamp, and the dataset includes some null or duplicate values. Although the dataset is rich, high-dimensional, and imbalanced, it can be used to train AI models to identify rare and advanced cyber threats.

3.2.3 Data Preprocessing

It is the third and vital portion to instruct a machine learning algorithm; the data set undergoes a preprocessing action, which determines the level of accuracy of the training model and the quality of information. The missing values are found and dropped to maintain the consistency of the column, and then duplicate rows are identified and removed to avert model overfitting and bias (Charbuty and Abdulazeez, 2021). The dataset contained numeric and categorical variables, which required converting the categorical labels, like Label, into numerical labels, and Label Encoding was used to do this.

```
[] df.duplicated().sum()
np.int64(5459)
```

Figure 6: Handle duplicate values

Moreover, columns that were not numeric, such as Timestamp, were removed since they were not useful in prediction and might have caused noise. Columns that had infinite values were changed to NaN, then imputed to avoid errors in the model. The data set was then divided into a training and testing set, with the class distribution maintained through stratified sampling. Lastly, SMOTE approaches were used to balance the classes by oversampling minority classes of attacks, such that the model is trained on both common and uncommon cyber threats.

EDA

Target Class Distribution

The distribution of target classes is visualised in this dataset, which is typical in understanding the imbalance, as it is a complex aspect in cybersecurity. The bar chart is plotted to show the number of samples in each class label; the labels are rotated and ordered by frequency for better

readability. It helps in detecting whether the attacks are present or not. If the dataset is imbalanced, then it leads to false detection.

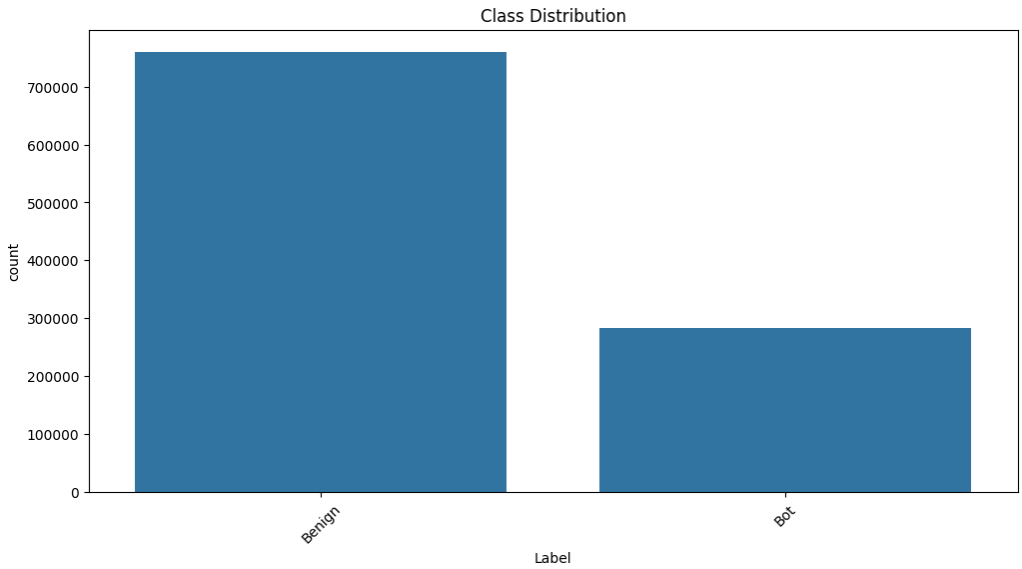

Figure 7: Class Distribution by Label

Feature Distribution

This presents a visualisation of the distribution of the feature Flow Duration according to the different class labels as a boxplot. Inside each box, the range, median, and spread of Flow Duration values are shown for a particular class (e.g., Benign, Infiltration, DoS). The duration of flows is broadly spread, hence a logarithmic scale (yscale('log')) is used to more clearly show the dispersion and identify extreme values or outliers. The plot helps in determining whether Flow Duration has the capability of differentiating between normal and attack traffic. The detection of such differences is valuable in the sense of choosing the relevant features when training a model and enhancing the model's capacity to identify cyber threats.

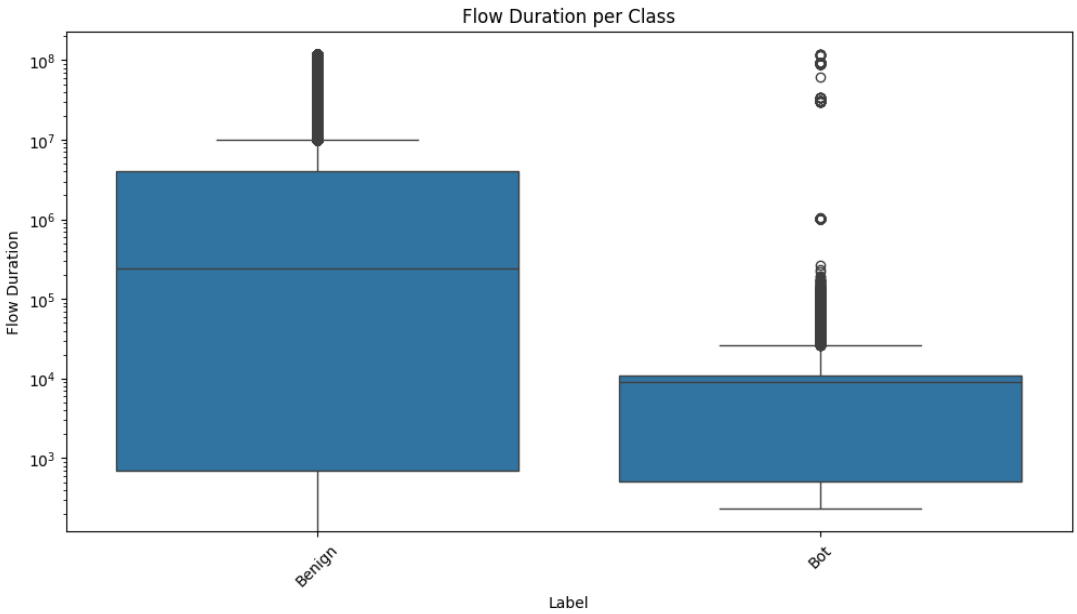

Figure 8: Flow duration per class

Protocol and Port Analysis

The relationship between protocol and port is analysed to understand the traffic trend in the dataset. Initially, it displays the frequency of every protocol, such as UDP, TCP, and ICMP, which shows which is used the most. After that, the top 10 destination ports are printed, which helps in detecting the target port over the traffic. Lastly, a bar plot (countplot) is created where Protocol is indicated on the x-axis, and Label is colored, indicating the association of benign or malicious activity with protocols. It is a process that allows identifying protocol-based attack patterns and allows the selection of features that can be used during model training.

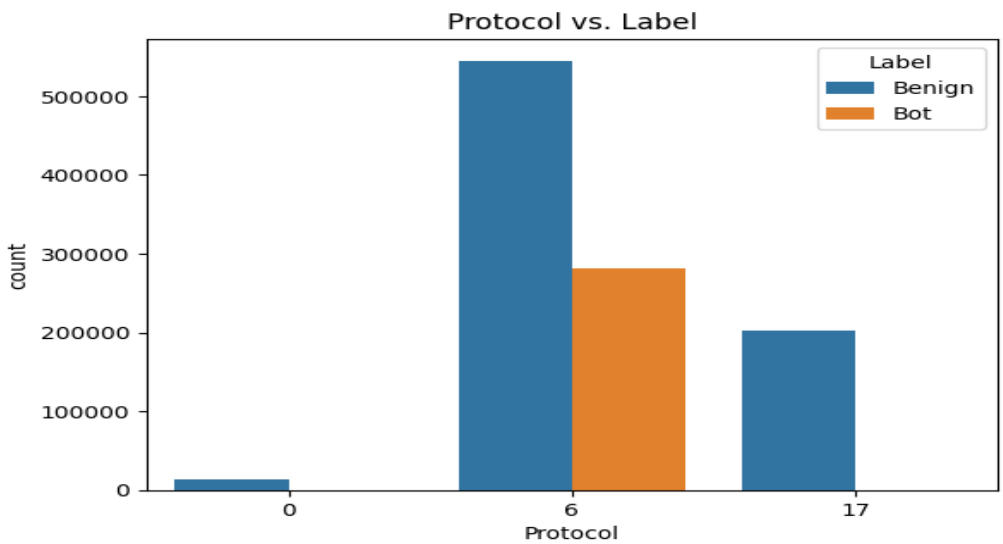

Figure 9: Protocol vs. Label

Top 15 Features by Variance

The top 15 features are identified which have the highest variance in this dataset. This feature gives useful information regarding machine learning approaches, through which the flexibility among the records is shown. Initially, it will compute the variance of all the columns containing numeric data (float64 and int64), and then take the top 15 results in descending order. A horizontal bar plot is then drawn to illustrate the above features and values of variance. The analysis helps in the selection of features in that it indicates which attributes are the most useful in identifying the difference between types of network traffic and can enhance the accuracy of the model to detect the threat.

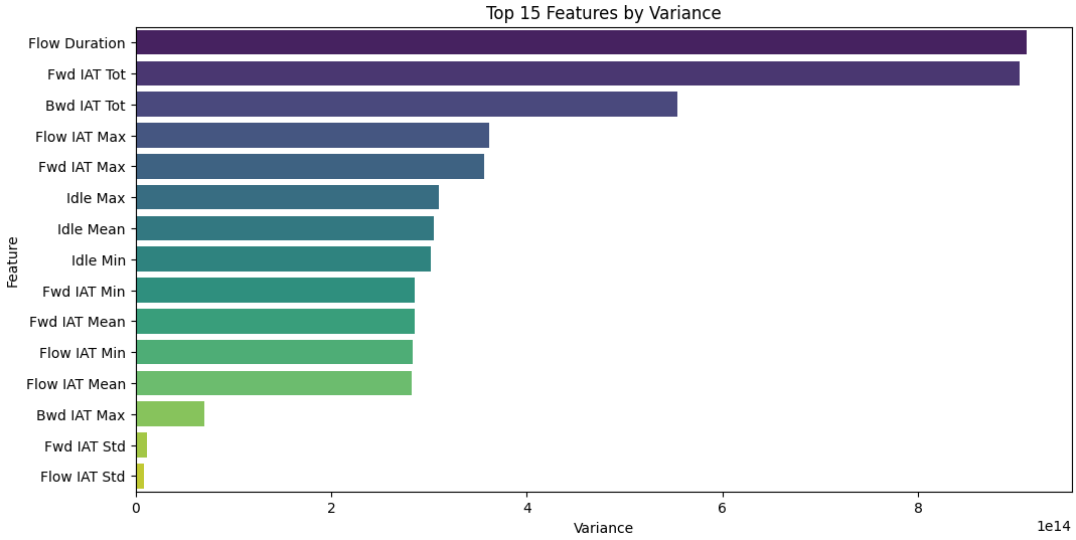

Figure 10: Top 15 Features by Variance

Pair-plot of key features by Label

A pairplot is generated that allows visualising the interrelationships among the selected numerical features and the target Label. To guarantee quick plotting, 1,000 rows are selected at random in the dataset. The chosen characteristics, Fwd Pkt Len Max, TotLen Fwd Pkts, Flow Duration, and TotLen Bwd Pkts, are plotted against each other, and each point is colored by the value of its class label. The diagonal displays the distributions of features with the help of KDE (Kernel Density Estimation). Such visualisation is useful to detect patterns, clusters or separation between normal and attack traffic, and then it is easier to understand what feature can be useful in classification tasks.

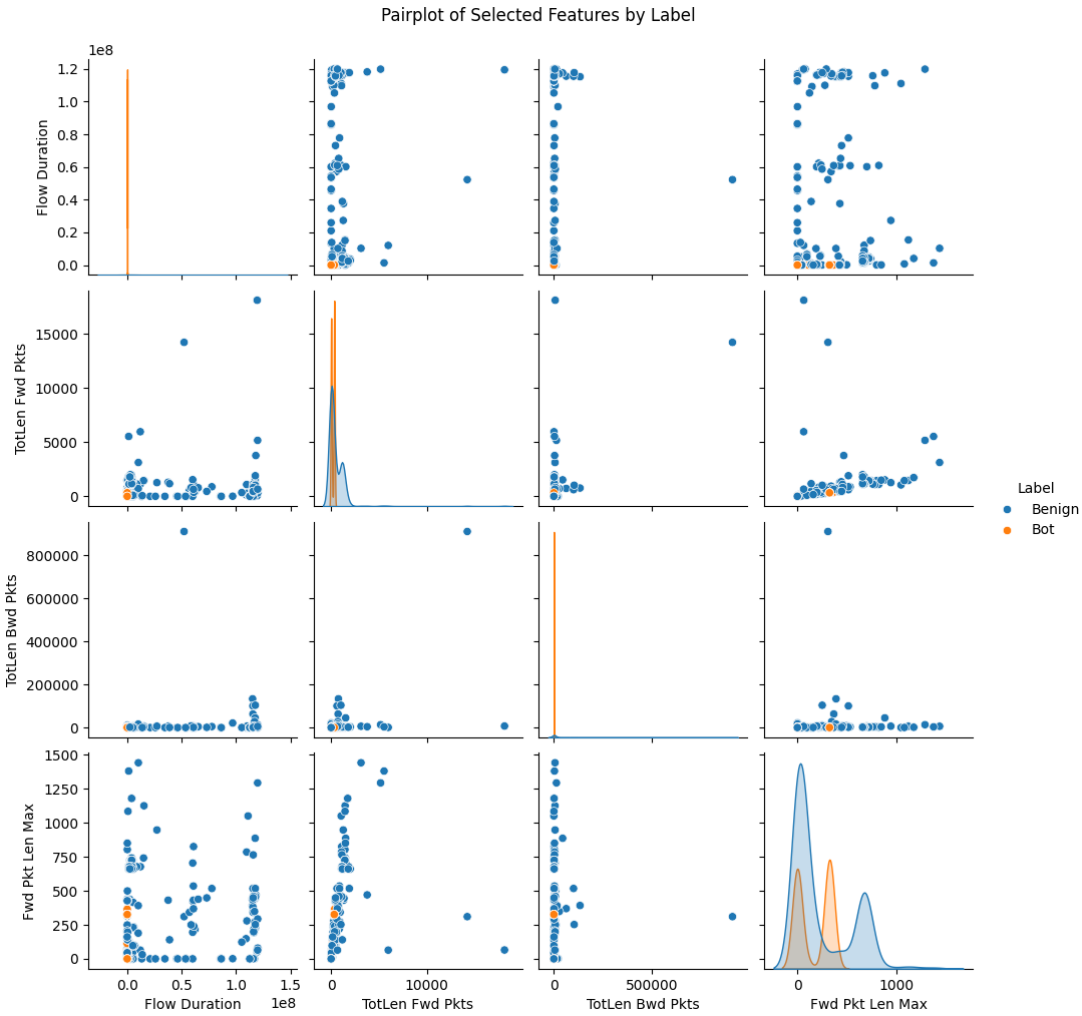

Figure 11: Pairplot of selected features by Label

Heatmap of Categorical Features vs Target

A heatmap is created to analyse the relationship among the top 10 target class labels and features. Initially, it converts the Dst Port column to a string along with the values of bin, except for the categorical label values. Then, it will form a cross-tabulation table whereby it will count the frequency of each port category with each class label. These counts are represented graphically in the heatmap, with the darker the colour, the more frequent the frequency. This analysis can be used to detect the possibility of some ports being correlated with some kind of attacks, which will assist in feature selection and attack behaviour.

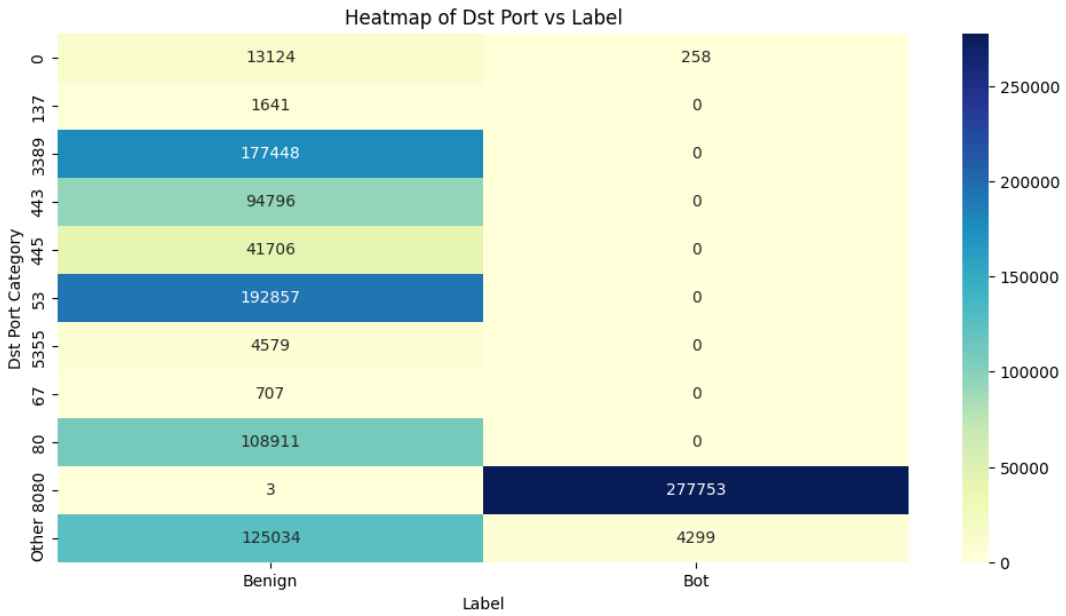

Figure 12: Heatmap of Dst Port vs Label

3.2.4 Modeling

Diverse schemes of detecting the cyber threat will be applied, including Decision Tree, Random Forest, KNN, etc. The future work is on the Random Forest boosting algorithm that trains multiple decision trees at once and combines the outputs of all the trees at test time. The classification issue is addressed quite successfully, and it comes in handy when working on imbalanced data, where it can handle both categorical and continuous variables, and it is able to estimate feature importance. Among the advantages, one should mention that Random Forest helps to avoid overfitting and is one of the most reliable and versatile models(Boateng *et al*., 2020).

The Decision Tree is a supervised learning method for categorising new data into subsets according to the feature values, thus creating a tree. Every internal node models a decision rule on an attribute, and every leaf node models one of the outcomes or classes. Decision Trees are very easy to explain and can manage both categorical and numerical data(Charbuty and Abdulazeez, 2021). As applied to attack category classification, Decision Trees have the advantage of partitioning the data into comprehensible decision paths in an endeavour to recognise attack patterns given by the feature splits.

KNN works through the use of distance between points, and categorising the input data point by the label of the k number of closest neighbours to the input data point (Boateng *et al*., 2020). However, KNN is operationally costly for large databases and adversely affects a noisy database. In this task, it will assist in determining the extent to which the attack categories

could be identified by proximity in the feature space, eventually giving an insight into the model's performance.

Logistic Regression has been used as a predictive analysis model and can be applied mainly to binary and multiclass classification. Probability is approximated through a logistic, and is most handy when there exists a linear relationship between the dependent and independent variables. When it came to attack category classification, the model was trained with a maximum of 1000 iterations so as to perform the necessary iterations for convergence (Chaudhary *et al.,* 2022). Based on the training process, predictions were made for the test set to evaluate this model by accuracy, confusion matrix, and classification report. Because of this simplicity, it had the ability to rapidly categorise the categories and still guarantee accuracy.

3.2.5 Evaluation

For evaluating the performance of the models, three primary metrics: confusion matrix, classification report and accuracy will be used (Vujović , 2021). These metrics gave the researcher a good insight into how well the attack categories were being predicted by the model.

Confusion Matrix

It gives a good method of calculating true positive, false positive, true negative and false negative measures in the classification model. This means that it will enable us to acknowledge the capacity that the developed model has in the classification between the various types of attacks(Krstinić *et al*., 2020).

Classification Report

The specific details of the confusion matrix are the precision, recall, and the F1-score for each class. Whereas precision is used to measure the ability of the model to find all and only the positives of a class, recall is the ability of the model to find all those that are relevant, and the F1-Score can be taken as the accuracy between precision and recall. The second analysis of the results is the more precise measure of the model performance on each of the kinds of attacks, so that it handles all types of attacks adequately without compromise.

Accuracy

Precision is a relatively uncomplicated measure that indicates the degree of correct prediction (the number of true positives and true negatives out of all total positive predictions). Although we have seen that accuracy gives an overall estimate of how a model is performing it should

not always be relied upon, particularly when the data set is not balanced(Agarwal *et al*., 2021). Hence, accuracy is reported along with the confusion matrix and classification report.

3.2.6 Deployment

The last phase is the deployment phase, which involves the implementation of the model that is going to be used to predict cyberattacks. It also enables the aspect to draw driven estimations that mainly depend on perceptions that are derived from the study. The deployment step is an implementation of a trained anomaly detection model in a real-time monitoring system. This makes a start by combining the selected model with the detection system, which makes sure that it can fetch up-to-date data for the prediction (Dixit *et al.,* 2021). Now, in the deployment phase, the model is used in a real-world environment where it can actively make a prediction. In this project, the model will be able to identify the attack types in real time, thus aiding the organisation to have an early picture of the threats. In order to keep the model optimised and working fine over time, the model will always be under observation. The deployment will provide an efficient and scalable intrusion detection solution, which can be explained as appropriate in terms of computing resources to provide support to small to medium-sized entities (Sarker, 2022).

3.2 Tools Used

Some of the features are operationalised through the use of the framework, and it is clear that they are utilised in the methodological approach of many application software, libraries, and programming languages, which can be helpful in analysing data and modelling (Kabir et al., 2024). Predominantly, the program in this is in Python and is also famous for their libraries and the frameworks that are important to machine learning and data science.

- **Pandas:** This is primarily due to the fact that the setting in which the pandas library can be analysed and modified is convenient data; it is also versatile in that after bringing out its analysis on the supplied data set, it can also alter the given data.
- **Matplotlib and Seaborn:** Some of the above visualisation libraries were used in the determination of some of the enhancements, fitting plots as well as charts, which we used while explaining different pros and cons of some aspects of data distributions, relations regarding the current status of the models of predictions, along with their future results (Dixit *et al.,* 2021).
- **NumPy:** It was described as a numeric processing library that also extends arrays for efficient data analysis, not only when used along with Pandas.

Chapter 4: Result Analysis

4.1 Introduction

Results analysis is a fundamental part of getting insight into the usefulness of machine learning models for any classification purpose. Several supervised learning techniques were used in the present research to improve network traffic detection by identifying and categorising it into Benign and Bot. There was an imbalance issue in the dataset arising between these classes, and using methods of addressing skew distributions in this dataset was therefore important. Before training the Model, the SMOTE method, Synthetic Minority Over-sampling Tool(SMOTE), was employed to rebalance the file. Four classification algorithms that are implemented are Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), and Logistic Regression (LR), following preprocessing and resampling.

The evaluation process was related to the determination of the precision, recall, F1-score, and accuracy of each class. To represent the classifications, confusion matrices were plotted for every model. Also, a comparative analysis based on the F1-scores to measure the model's suitability for the detection of bots was conducted.

4.2 Data Preparation and SMOTE Application

The dataset was thoroughly preprocessed prior to successfully training models so as to create the best performance. Timestamp column and any other attribute of type non-numeric were excluded, since they did not themselves have any direct predictive power. The infinite values were set to NaN, and then they had to be filled with column means to prevent data loss.

```
SMOTE applied successfully!
New class distribution:
 Label
Bot       608644
Benign    608644
Name: count, dtype: int64
```

Figure 13: SMOTE Classification

Since the dataset was divisible, the SMOTE technique was used to create artificial samples of the minority class. The resampling operation was able to balance both classes, where Benign had 608,644 records and Bot had 608,644 records. This made sure that classification algorithms were not biased against the majority class and had more ability to identify malicious patterns of traffic.

4.3 Random Forest

The high accuracy rate indicated by the random forest algorithm, a machine learning technique that employs numerous decision trees and combines their determinations, implied that it had an ideal rate of classification in the given analysis. When innocent traffic was considered, precision, recall, and F1-score referred to the model of 1.00, i.e. all the non-malicious flows of the network could be detected. The same scenario was observed with respect to bot traffic, with the model attaining perfect scores across all the metrics, indicating its performance to be able to identify all the malicious cases correctly without misclassifying the benign ones. The results achieved an overall accuracy of 1.00, which implies that all cases within the test set got their classification.

```
Random Forest Classification Report:
              precision    recall  f1-score   support

      Benign       1.00      1.00      1.00    152162
         Bot       1.00      1.00      1.00     56462

    accuracy                           1.00    208624
   macro avg       1.00      1.00      1.00    208624
weighted avg       1.00      1.00      1.00    208624
```

Figure 14: Random Forest Classification Report

These findings were corroborated by the confusion matrix, since all of the samples belonging to either class were correctly predicted. With such a high score, Random Forest may have picked up the unique features of both types of traffic, benign and bot. Nonetheless, one should mention the fact that such large results can be noted as being suspicious of overfitting, especially in cases where synthetic oversampling procedures such as SMOTE are applied. There are two prospects associated with the generated synthetic data in such cases: firstly, it can be very similar to the original samples, and therefore, the model can fit perfectly high scores, but it can also decrease its capability to generalise in real-world, unseen data.

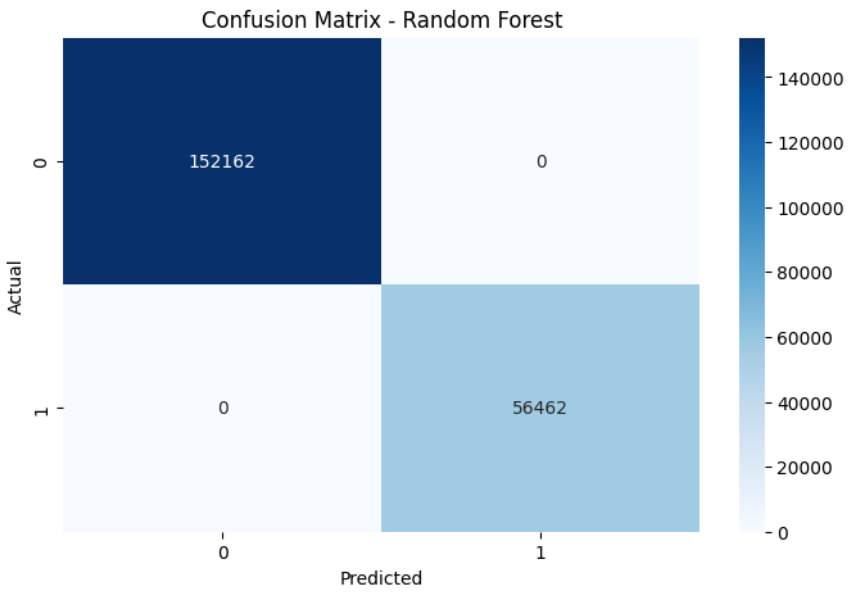

Figure 15: Confusion Matrix Random Forest

4.4 XGBoost

XGBoost, an effective gradient boosting framework, which is well-known to be efficient when processing structured datasets, provided a similarly accurate result when it came to classification performance in this analysis as well. The model that operated on Class 0 (Benign) had the precision, recall, and F1-score of 1.00, which identified all the false traffic without misclassification as the valid network flows. In a similar manner, the precision, recall, and F1-score of Class 1 (Bot) were also ideal and to the tune of 1.00, which indicates that all the malicious cases were properly identified without being identified as false positives. The general precision was 1.00 and proved that all test samples were accurately categorised. The results were also supported in the confusion matrix, where no classifications were in error with each other.

```
XGBoost Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    152162
           1       1.00      1.00      1.00     56462

    accuracy                           1.00    208624
   macro avg       1.00      1.00      1.00    208624
weighted avg       1.00      1.00      1.00    208624
```

Figure 16: XGBoost Classification Report

This outstanding result can be explained by a variety of reasons, such as the balanced dataset, which was generated by SMOTE, and the very good separability of features between benign

and bot traffic. Also, the gradient boosting process implemented by XGBoost, which sequentially builds models and corrects on each iteration, may be the critical determining factor in such high levels of accuracy. Nonetheless, as is the case in the results of Random Forest, the idealised classification suffers from a similar issue that the model might have memorised patterns unique to the dataset used to train it, which can limit its applicability to the general aspects of any real-world data entirely unseen to the training dataset.

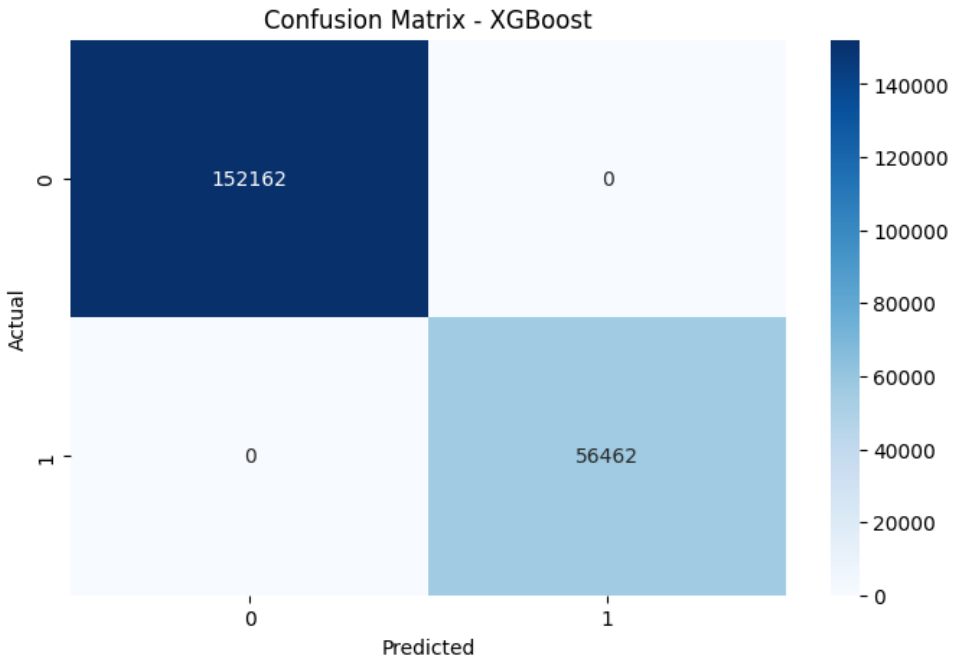

Figure 17: XGBoost Confusion Matrix

4.5 Support Vector Machine (SVM)

The results of the Generalised Linear Models featured a Support Vector Machine (SVM) classified with a linear kernel, and the accuracy level was also high, but lower than within the ensemble-based models. The benign traffic model has been precise by 100 per cent, recalling 95 per cent and an F1-score of 97 per cent, as none of the benign instances were classified as malicious, but a small percentage of actual benign instances were overlooked in the classification. In the case of bot traffic, the model was able to achieve a precision of 0.88, perfect recall of 1.00 and F1-score of 0.93, implying that, out of all the instances predicted as malicious, the model was able to identify correctly all the malicious instances at the expense of some false positive cases where benign traffic was incorrectly labeled as malicious bot traffic.

```
SVM Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.95      0.97    152162
           1       0.88      1.00      0.93     56462

    accuracy                           0.96    208624
   macro avg       0.94      0.97      0.95    208624
weighted avg       0.97      0.96      0.96    208624
```

Figure 18: SVM Classification Report

The SVM model had an overall accuracy of 0.96, which was also very high, but not as high as Random Forest and XGBoost. This difference in performance could be because a linear decision boundary may not be flexible enough to have non-linear relationships that may exist in the feature space. Nonetheless, the infallible bot detection is what makes SVM relevant in instances of cybersecurity where there is an absolute need to minimise false negatives, since the loss of malicious traffic may carry graver impacts than misidentifying a harmless piece of information.

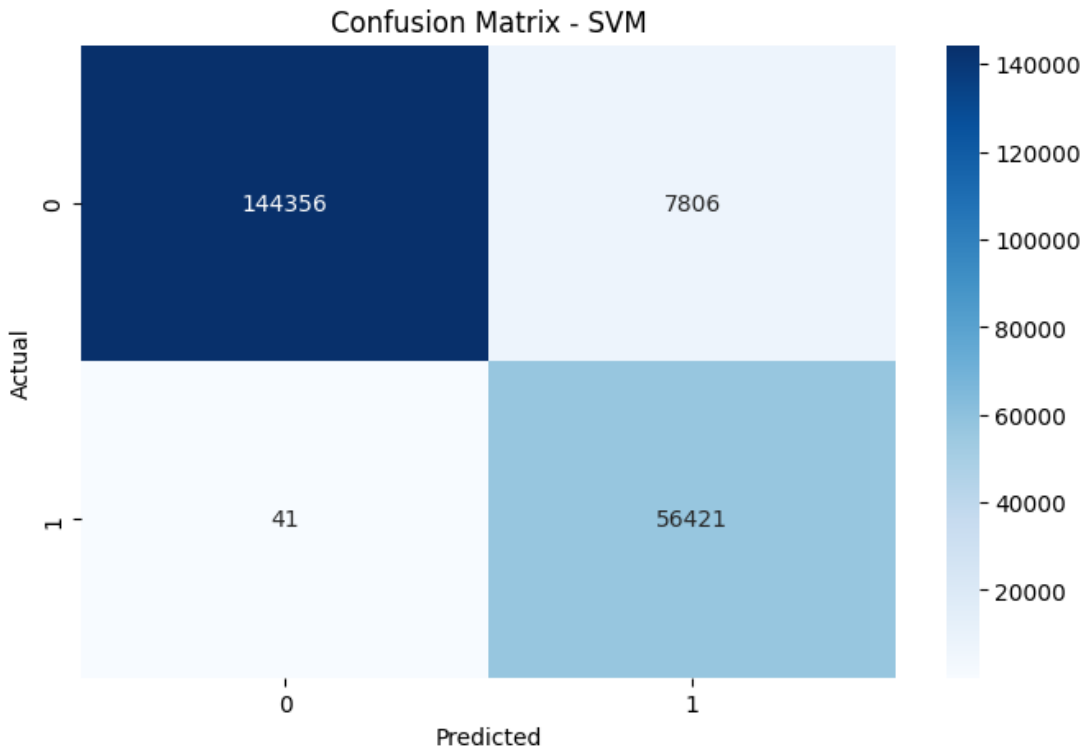

Figure 19: Confusion Matrix for SVM

4.6 Logistic Regression

Common statistical classification method-Logistic Regression performed competitively but was slightly outperformed by the ensemble-based models in the aggregate accuracy. On benign traffic, the model obtained a precision of 1.00, a recall of 0.97 and an F1-score of 0.98, meaning that although the model correctly classified malicious, one per cent of actual malicious data were confused with benign. To the bot traffic, the model achieved a precision of 0.91, a recall of perfect 1.00 and the F1-Score 0.95, which acts as substantiation to the fact that the model was able to identify all the malicious traffic correctly, but also at times was unable to accurately identify all the benign traffic. The total precision was 0.97, and this indicates a high ability to predict.

```
Logistic Regression Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.97      0.98    152162
           1       0.91      1.00      0.95     56462

    accuracy                           0.97    208624
   macro avg       0.96      0.98      0.97    208624
weighted avg       0.98      0.97      0.97    208624
```

Figure 20: Logistic Regression Classification Report

As the confusion matrix indicated, LR was also substantive in categorising bot traffic, as not a single case was incorrectly identified as such. Partly, this good performance can be attributed to the fact that the data scaling was performed at the stage prior to training, and all features obtained an equal role in the data classification process, and those with a greater numerical range did not overweight the results in the decision-making algorithm of the model. Despite the slightly lower performance with respect to Random Forest and XGBoost, Logistic Regression turned out to be a stable tool in the area of network traffic classification, especially where the high value of recall associated with bot identification is of primary concern.

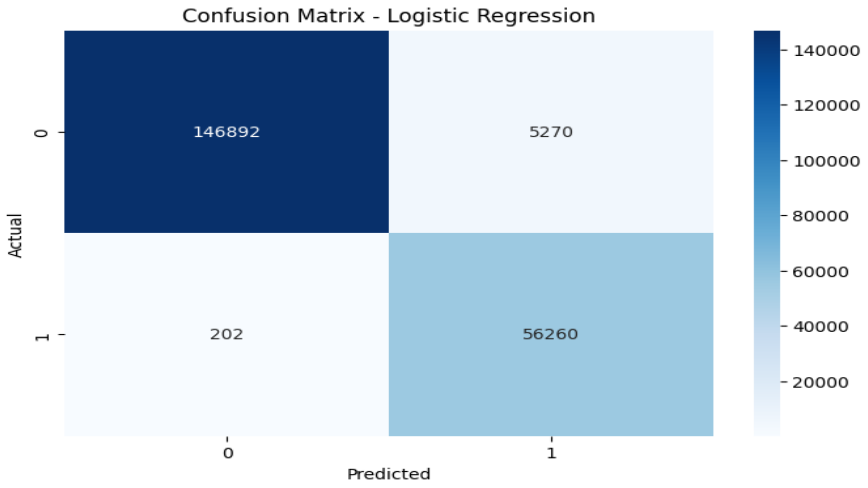

Figure 21: Confusion Matrix for Logistic Regression

4.7 Confusion Matrix Interpretation

The confusion matrices demonstrated the performance of each model in a simple and easy-to-understand manner, indicating the classes of benign and bot traffic. Random Forest and XGBoost showed an optimal result as there were no cases of misclassification; thus, the predicted label was perfectly matched with the observed label in both classes. In comparison, the Support Vector Machine (SVM) and Logistic Regression models had slight errors, with a majority of them being benign samples that are classified as bot traffic. The trend depicts a typical trade-off regarding classification challenges when there is a need to minimise false negative error, when the malicious traffic may not have been found, against the minimisation of the false positive error, where the legitimate traffic can wrongly be detected as the malicious traffic.

To the security context of the network, the ramifications of an all but malicious loop being missed are typically viewed as more severe than the nuisance of overly labelling unthreatening activity. This is why the models with perfect recall of bot traffic, like XGBoost and Random Forest, gain certain importance. The models have the advantage that any malicious examples are guaranteed to be detected and thus reduce the chance of having undetected threats, though at the possible cost of a small increase in the false positive rate of other models.

4.8 Comparative Analysis of Models

The difficulty of comparing the F1-scores in terms of visual presentation provided a clear and concise visual representation of the performance of the four tested models. Although Random Forest and the XGBoost had perfect F1-scores on both classes, benign and bot traffic, it is important to note that they have been shown to be strong and to perform with high accuracy in

both cases, resulting in a high F1-score. The Support Vector Machine (SVM) showed a slight decrease in performance on benign traffic, mostly because of lower recall, but the overall performance of bot detection was comprehensive, especially in terms of perfect recall on the particular class.

Figure 22: Model Comparison

One area where Logistic regression performed impressively during F1-scores was bot traffic, but underperformed slightly with regard to benign traffic, just as with the ensemble models. Compared on the key indicator of the overall accuracy, recall and F1-Score, one would notice that the ensemble-based models Random Forest and XGBoost provided much stronger results than the linear models, including SVM and Logistic Regression. This high performance shows the benefit of ensemble methods in identifying non-linear, complex relations with the data, which has the capability to give more exact and durable classifications in network traffic analysis.

4.9 Chapter Conclusion

Analysis of results demonstrates that ensemble methods (Random Forest and XGBoost) are capable of reaching remarkable performance in bot traffic detection under the condition of well-balanced data. There was also a very good performance with slightly lower rates of accuracy in SVM and Logistic Regression, since they misclassified benign traffic.

SMOTE was effective in solving the issue of class imbalance, and all models had a high score in terms of recall of Bot traffic. Random Forest and XGBoost can be stated as the most applicable of the analysed models to be applied to the situations where detecting any malicious activity with the minimal number of false negative results is essential.

Nevertheless, the overall good results, especially the accuracy of 100 per cent in the models, make it worthwhile to consider additional testing with more varied and realistic data. The comprehensive assessment strategy that would include the cross-validation data and external validation data would make sure that the models are capable of performing in real-world settings.

All in all, this analysis proves that, when meticulously preprocessed and balanced, data can make machine learning models work at near-perfect levels, yet it also goes on to emphasise the need to test these findings outside the training environment.

Chapter 5: Conclusion and Recommendation

5.1 Conclusion

The paper attempted to discuss the role of Artificial Intelligence (AI) and Machine Learning (ML) in enhancing predictive cybersecurity threat intelligence with an emphasis on identifying uncommon yet hazardous cyber-attacks. Based on the CIC-IDS2018 dataset, the study tackled one of the typical problems in cybersecurity datasets, namely class imbalance, where network traffic is ordinary, and a considerably minute fraction represents the attacks. This kind of imbalance may lead to the failure of classical models to detect minority-class threats, such as web attacks, infiltration, or advanced persistent threats (APTs), which, though low, may cause devastating effects.

The analysis returned to the Synthetic Minority Oversampling Technique (SMOTE) to produce additional samples of minority classes in order to overcome this problem. This allowed the models to better train on the features of rare attacks. In this paper, four algorithms were tested: Random Forest (RF), XGBoost, Support Vector Machine (SVM), and Logistic Regression (LR). Precision and recall, F1-score, and confusion matrices were employed to estimate the performance, and in particular, recall was of particular interest because a false negative (attack is missed) represents a more critical risk in terms of cybersecurity than a false alarm.

The findings indicated an accuracy, precision, recall, and F1-score of 100 per cent in both the Random Forest and XGBoost running on the balanced dataset to identify all the benign and malicious traffic without making an error. SVM and Logistic Regression also showed good results, particularly in identifying malicious traffic, but they incurred small errors in failing to identify some benevolent traffic as malicious. That implies that they were a little less precise on benign traffic; however, they kept a perfect recall on bot traffic, which is of great importance to security operations.

The overall performance of the study was that, when taking appropriate preprocessing, class balancing, and properly selecting the model, machine learning can attain highly optimistic performance in defining the minority-class cyber threats. Nevertheless, it would be necessary to see it in the real-world setting, under which examinations of live, or more diverse data must be utilised to avoid the possibility of merely having a limited result of the training scenario.

5.2 Recommendations

Depending on the results, the following recommendations are offered for future research and practice:

Real-World Data test

The models worked perfectly on the existence of a balanced dataset; however, real network traffic is highly complex and noisy. Future research needs to prove the efficacy of the models on real-life, continuously updated data to make sure that it can generalise on a larger scale than the controlled testing conditions did.

Apply Hybrid resampling Methods.

Despite the effectiveness of SMOTE, the implementation of additional methods, such as ADASYN or SMOTE-ENN, may have further benefits in terms of model robustness, as those approaches may further eliminate the risk of noise and overfitting.

Continuous Model updating

Threats in the cyber world keep changing. Use in production must consist of regularly re-training with new data to reflect new patterns of attack, along with retention of high detection rates.

Include tools of Explainability.

Security analysts have to be confident in the decisions of the system. With explainable AI (i.e., SHAP or LIME), it will be possible to interpret why a particular traffic pattern was identified as malicious and enhance trust and clarity among the users.

Quality Recall / Precision Trade-off in Practice

Although high recall proves vital, especially toward the identification of all the threats, the number of false positives might overwhelm security teams. In the future, it is unlikely that an

optimal tuning is possible, which would minimise false negatives and generate few false alarms.

5.3 Final Statement

This paper confirms that machine learning can be applied with a high degree of success to detect rare cyber threats with a specific, efficient method of preprocessing and balancing data. The ensemble models, such as Random Forest and XGBoost, have great potential, yet some care should be taken in order to make sure they continue to work within real-world, dynamic cybersecurity scenarios. As these AI-powered models continue to be tested, explainable, and become part of operational security systems, there is potential that they will become an invaluable asset to up-to-date cyber defence operations.

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